Lesson plan · for the teacher

Guardrails: rules around the model

AI security · Level 2 · Lesson 3

Time
40 min

Materials
A card with five banned words for the “robot”, such as homework, answers, copy, test, cheat; Slips of paper for the messages; Board and chalk for the tally

Learning goals

- Describe three kinds of guardrail around an AI model.
- Explain why word lists miss rewording and block innocent text.
- Explain why a check on the action holds.

Activity: The word-list robot

1. Warm-up (5 min): Ask how the school gate decides who comes in: a list of people who are banned, or a pass that people must have? Which works better for a stranger?
2. Teach (8 min): Explain the three guardrails around an AI: a filter on what goes in, a filter on what comes out, and a check on what it is about to do. Show why a list of words misses rewording.
3. Round 1 (12 min): The robot holds the card of five banned words and stops any slip that uses one. Classmates reword to get past. Tally what got through, and any innocent slip that was stopped.
4. Round 2 (10 min): Now every slip needs your yes before it is passed on. Play again and compare the two tallies.
5. Wrap-up (5 min): Agree that checking the action works whatever the wording, and that the best defence is layers. Then give the exit ticket.

The messages ask only for homework answers, nothing harmful. Watch for innocent slips the robot stops: they are the false alarms worth discussing.

Discussion

- Why did reworded messages get past the robot?
- Which innocent message did the robot stop?
- Why did the teacher's yes work whatever the wording?

Exit ticket

- Write one weakness of a word filter, and one guardrail that holds whatever the wording.

Try it at school tomorrow
Build a “word-list robot”: one pupil blocks messages with banned words, classmates reword a harmless message to get past, then a teacher must approve every message. Compare what got through each way.

Practice questions · for you

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1 **Which guardrail stops a message nobody approved, however it's worded?**

- ◯ A longer list of banned words
- ◯ A check in code that it was asked for and approved
- ◯ An output filter in English only

2 **Match each guardrail to what it checks.**

Input filter	• • • •	Whether this action was asked for and approved
Output filter	• • • •	What the person types
Action check in code	• • • •	What slipped past all the others
People and reports	• • • •	What the model is about to say

3 **A good word filter blocks harmful messages without ever blocking innocent ones.**

- ◯ True
- ◯ False

4 **Put the layers in the order a message meets them.**

- ☐ The output filter checks the message
- ☐ The model writes a message
- ☐ Code checks it was approved, then sends
- ☐ The person types a request
- ☐ The input filter checks it

5 **Which of these make a school chatbot safer as layers? Choose all that apply.**

- ☐ An input filter
- ☐ People who read reports of problems
- ☐ A check in code before any action
- ☐ Trusting the model never to make mistakes

✎ **Think of an AI tool at your school, or one you might choose. Which guardrails would you ask the provider about, and which action would you want a person to approve?**

Check your answers

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1. **A check in code that it was asked for and approved**

   A longer list still misses new wordings and other languages.

2. **Input filter — What the person types**
   Output filter — What the model is about to say
   Action check in code — Whether this action was asked for and approved
   People and reports — What slipped past all the others

   Each layer looks at a different moment: the request, the reply, the action, and afterwards.

3. **False**

   Word filters make both mistakes: they miss reworded attacks and block innocent text, such as a lesson that mentions passwords.

4. **Top to bottom: 4, 3, 5, 1, 2**

   A request is filtered on the way in, the model writes, the reply is filtered on the way out, and the last gate before anything happens is the check in code that a person approved it.

5. **An input filter; A check in code before any action; People who read reports of problems**

   Filters, action checks and people each catch some of what the others miss.